



Aditya Aryan

Bachelor of Statistics (Hons.)
Indian Statistical Institute, Kolkata

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Education

Indian Statistical Institute

Bachelor in Statistics (B.Stat), First Division with Hons.(Projected)

Jul. 2023 – Present

Kolkata, West Bengal

Kendriya Vidyalaya No. 2, Jalahalli East

12th CBSE Boards (PCM) - 95%

10th CBSE Boards - 98.8%

Apr. 2014 – Mar. 2023

Bangalore, Karnataka

Relevant Coursework

- Real Analysis
- Probability Theory
- Stochastic Processes
- Statistical Methods
- Linear Algebra
- Differential Equations
- Programming & Data Structures
- Molecular Biology
- Psychology

Experience

Student Researcher

University of Toronto, Advisor: Prof. Xin Bing

Apr. 2026 – Present

Remote

- **Project:** Studying convergence properties of the Expectation-Maximization algorithm for Gaussian mixture-type models using tools from high-dimensional probability and statistical learning theory.
- **Problem:** Analyzing when population and sample EM iterates contract toward the true parameter under non-convex likelihood geometry, mixture imbalance, and high-dimensional noise.
- **Methods:** Working with population EM maps, fixed-point analysis, local contraction arguments, Stein's lemma, Fisher/missing-information decompositions, and concentration-style high-dimensional probability bounds.

Data Science Intern

Ranial Systems

May, 2026 – Present

Remote

- **Project:** Developing machine learning and optimization models for U.S. Battery Energy Storage Systems (BESS), with applications to operational decision support, performance monitoring, and energy trading optimization.
- **Modeling:** Building end-to-end ML workflows for BESS optimization tasks, degradation/performance analysis, alert-level prediction, survival modelling and models for Energy-market optimisation.
- **Implementation:** Creating reproducible Python based pipelines, evaluation dashboards, and decision-support outputs to translate raw BESS telemetry and market data into decision making models.

ML Research Intern

Mercy AI

Apr – Aug 2025

Bengaluru, Karnataka

- **Project:** Compact vision-language model for medical image reasoning; integrated medSigLIP (vision encoder) with Qwen-2.5-7B via Q-Former, compressing 256 image tokens to 32 latent queries ($8\times$ compression, 70% memory reduction) while preserving probabilistic calibration in frozen-LLM caption generation.
- **Methods & Outcomes:** Trained Q-Former with cross-entropy on MIMIC Captions (frozen encoder and LLM); explored SFT, PPO, DPO, and GRPO via Hugging Face `trl`; extended DeepSeek-R1-based GRPO pipeline for multimodal reasoning; BLEU improved and perplexity decreased; published reproducible training configs and technical blog posts.

Projects

Stein's Paradox: Inadmissibility and Risk-Optimal Shrinkage

Supervisor: Dr. Ayanendranath Basu, ISI

- **Problem:** Provide empirical evidence for Stein's paradox by demonstrating the inadmissibility of the MLE for multivariate normal mean estimation ($d \geq 3$), and characterize risk improvements from shrinkage estimators under squared-error loss.
- **Methods:** Developed a risk-first Empirical Bayes framework; derived James-Stein and mean centered shrinkage; extended to heteroscedastic normal means via whitening and Normal-Normal Empirical Bayes (MoM and MMLE); evaluated risk using Monte Carlo simulations and Stein's unbiased risk estimate (SURE).
- **Results:** Empirically confirmed uniform risk dominance of James-Stein over MLE; showed MMLE achieves lowest risk in heteroscedastic regimes, while MLE is consistently suboptimal; results matched classical decision-theoretic predictions.

Bivariate Copula Modelling of Extreme Air Pollution Events

Supervisor: Dr. Shyamal Krishna De, ISI

- **Problem:** Model the joint distribution of duration D and cumulative severity S of extreme AQI episodes in Bangalore (2015–2024) without imposing parametric marginal assumptions.
- **Methods:** Extracted unhealthy events ($\text{AQI} > 100$) from a decade of hourly data; constructed average-rank pseudo-observations for discrete D ; fit six bivariate copulas via canonical MLE; selected by AIC; derived exact conditional exceedance probability $P(S > s_0 \mid D \geq d_0)$ using empirical marginals.
- **Outcomes:** Confirmed extreme upper-tail dependence ($\hat{\tau} = 0.881$, $\hat{\rho}_S = 0.972$); Gumbel copula selected ($\hat{\theta} = 6.183$, $\hat{\lambda}_U = 0.881$); risk assessment yielded $P(S > 2000 \mid D \geq 12) \approx 0.9998$, establishing a 12-hour duration as an actionable public-health escalation threshold.

Sequential Testing: Optimal Stopping under Reward Cost Tradeoff

Supervisor: Dr. Arnab Chakraborty, ISI

- **Problem:** Maximize expected cumulative gain from sampling N boxes sequentially, where C boxes contain rewards of value V , each sampled box incurring some cost, and reward locations are unknown.
- **Methods:** Modeled the process as a Markov decision problem with game state $S(x, y)$ representing x rewards and y non-rewards discovered. Formulated dynamic programming recursion for maximum expected gain $\text{GAIN}(x, y)$ at state $S(x, y)$; implemented exact memoized DP and backtracking algorithm; constructed cutoffs using $\text{GAIN}(x, y)$ to define optimal strategy.
- **Outcomes:** Derived and proved the recursive optimal strategy; implemented memoized recursion and backtracking in C++; computed exact expected gain for $N = 20, C = 5$ as 7.6875; result matched with theoretical optimal strategy.

Variance-Based Audio De-Noising via Amplitude Thresholding

Supervisor: Dr. Arnab Chakraborty, ISI

- **Problem:** Remove noise from audio recordings using statistical detection of silent or low-variance intervals without frequency filtering.
- **Methods:** Computed amplitude variance over sliding windows (length ≈ 0.1 sec); normalized and binned variances into discrete classes; used modal variance bin to define adaptive cutoff; suppressed or scaled amplitudes in low-variance regions; implemented in R with FFT diagnostics for validation.
- **Outcomes:** Successfully removed background noise in clinical and synthetic samples; preserved speech intelligibility in overlapping frequency bands; visualized filtered waveform and FFT; pseudocode and full implementation included.

Elo-Based Football Betting

Supervisor: Dr. Ayanendranath Basu, ISI

- **Problem:** Design a statistically calibrated, profitable football match betting strategy using historical match outcomes and odds.
- **Methods:** Computed team Elo ratings updated matchwise; modeled match outcomes using logistic regression over Elo-difference features; calibrated model probabilities; evaluated using Brier score and log-loss; developed value-betting rule using market odds and model probabilities; backtested on 2014–2024 EPL data.
- **Outcomes:** Logistic calibration improved probabilistic accuracy (e.g., Brier 0.2176, Log-loss 0.6249); betting simulation yielded total simulated profit of 5,139 currency units on 10,000 bankroll over test season; plotted calibration curves and performance metrics. Code, data, and evaluation script provided.

Skills

Mathematics & Statistics: Probability theory, Markov Processes, High dimensional statistics, hypothesis testing, linear modelling, time-series modelling.

Programming: Python (advanced), R (intermediate), PyTorch, scikit-learn, statsmodels, pandas, NumPy, SciPy; Hugging Face Transformers: ‘trl’, ‘accelerate’, ‘bitsandbytes’, ‘peft’.

Domains: Machine learning, Transformers, Vision Transformers, CNNs, diffusion models, Deep Reinforcement learning, Computer Vision, Language modeling.

Scholastic Achievements

Secured All India Rank 90 in ISI entrance examination.	2023
Secured 98.9%ile in JEE Mains(B.Tech) among 1.5 million candidates.	2023
Scored 331/390 in BITSAT.	2023
Secured All India Rank 1034 in KVPY(SA stream).	2022
Qualified for INMO (Indian National Mathematics Olympiad) merit list.	2022
Secured a Top 20 Rank in IOQM in Karnataka region.	2022
Received NTSE scholarship for being in top 1000 students among 2 million candidates.	2021
Secured Rank 72 in NTSE Stage-I.	2021